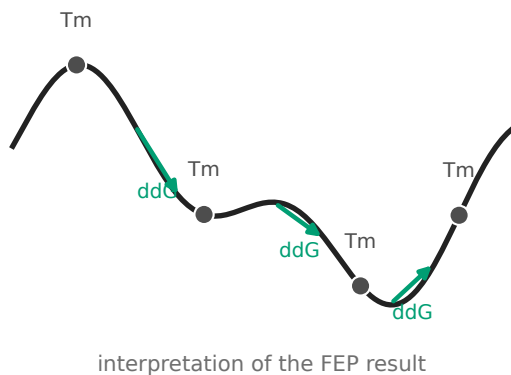
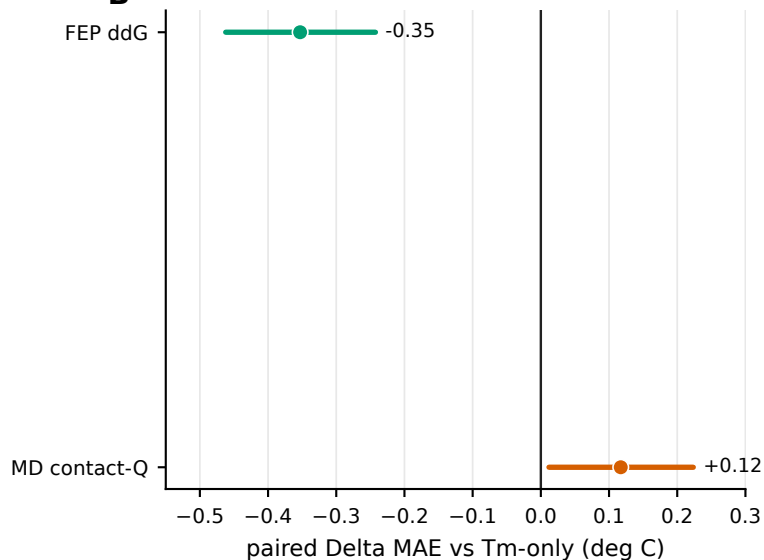


A

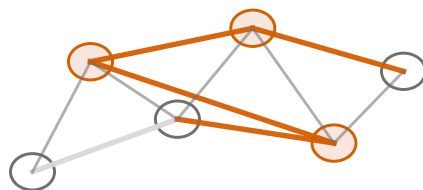
Why mutation-effect labels can help absolute anchors + mutation effects

**B**

Mutation-effect vs MD-derived label

**C**

Contact-Q definition



Q = retained native hydrophilic contacts
400K trajectory, final 30 ns average

D

Interpretation

FEP labels encode mutation effects that are close to stability perturbations.

MD contact-Q summarizes structural persistence; it did not improve final Tm prediction here.

Take-home: useful simulation labels must carry information relevant to Tm generalization.